

Materials for Printing and

□ To provide basic exposure to the common raw materials and consumables used in

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3102 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3102.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Printing Technology
Course code	3102
Course title	Materials for Printing and
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45

Item	Official information
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

22 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To provide basic exposure to the common raw materials and consumables used in printing and packaging practices.
- □ To guide the student to know about various substrates and other essential materials used in the printing and packaging industry.
- □ To understand the general concepts on characteristics, manufacturing and uses of primarily used raw materials in printing and packaging.
- □ To understand the role of primarily used raw materials in printing and packaging in deliver quality products at the output.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding Summarize the various aspects of paper and	See official syllabus
CO2	12 Understanding board used in printing and packaging. Explain about various aspects of plastics used	See official syllabus
CO3	10 Understanding in printing and packaging. Describe about various aspects of	See official syllabus
CO4	9 Understanding miscellaneous printing materials.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2

Row	Extracted official mapping
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe about different aspects of printing inks.	4	As specified
Module 2	packaging.	7	As specified
Module 3	Explain about various aspects of plastics used in printing and packaging.	5	As specified
Module 4	Describe about various aspects of miscellaneous printing materials.	6	As specified

MODULE 1

Describe about different aspects of printing inks.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe about different aspects of printing inks.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Ink-introduction and components

Ink-introduction and components is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Ink-introduction and components using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ink-introduction and components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Ink-introduction and components means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇൻക്-ഇൻട്രോഡക്ഷൻ ആൻഡ് കമ്പോണൻ്റ് ഇൻക്-ഇൻട്രോഡക്ഷൻ
ഇൻക്-ഇൻട്രോഡക്ഷൻ definition/meaning ഇൻക്-ഇൻട്രോഡക്ഷൻ. ഇൻക്-ഇൻട്രോഡക്ഷൻ steps,
application ഇൻക്-ഇൻട്രോഡക്ഷൻ ഇൻക്-ഇൻട്രോഡക്ഷൻ. Technical terms English-ഇൻക്-ഇൻട്രോഡക്ഷൻ
ഇൻക്-ഇൻട്രോഡക്ഷൻ.

Ink properties

Ink properties is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Ink properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ink properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Ink properties means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇൻക് പ്രകൃതികൾ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. ഇൻക് പ്രകൃതികൾ, steps, application നിർവ്വചിക്കുക. Technical terms English-ഇൻക് പ്രകൃതികൾ.

Ink manufacturing

Ink manufacturing is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Ink manufacturing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ink manufacturing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Ink manufacturing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇൻക് മാനുഫാക്ചറിംഗ് നിർമ്മാണത്തിന്റെ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർമ്മാണത്തിന്റെ ഘട്ടങ്ങൾ, steps, application നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and

Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, types of printing inks 3 understanding ink- introduction, raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. ink properties: optical properties - color, gloss, opacity, transparency. physical properties - length, viscosity, tack, flow, adhesion, scuff resistance. rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. ink manufacturing - flow chart, illustration and description of- varnish manufacture, pigment dispersal, mixing, milling, filtering, quality check, packing and labeling. name of standard ink manufactures in india and world. types of printing inks: offset inks- non-heatset (coldset) inks, heatset inks, quickset inks, sheet-fed inks, gravure inks, flexo inks, screen-printing inks, water-

based inks, laser ink, uv ink, process ink colors, high- fidelity ink. specialty ink- non-porous ink, metallic ink, fluorescent ink, magnetic ink, scratch and sniff ink, edible ink, moisture resistant ink, security ink. common ink related printing problems and their remedies-set off, summing, hickies, picking, linting, spot, mottle, strike through. summarize the various aspects of paper and board used in printing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset

Official syllabus detail and terminology

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 2

packaging.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: packaging.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,

2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding raw materials used. pulping-pulp washing, bleaching, refining, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാഠ്യ 2 Understanding raw materials used. Pulping- Pulp Washing, Bleaching, Refining, പാലപ്പൊട്ടിപ്പം പാലപ്പൊട്ടിപ്പം definition/meaning പാലപ്പൊട്ടിപ്പം. പാലപ്പൊട്ടിപ്പം പാലപ്പൊട്ടിപ്പം, steps, application പാലപ്പൊട്ടിപ്പം പാലപ്പൊട്ടിപ്പം. Technical terms English-പാലപ്പൊട്ടിപ്പം പാലപ്പൊട്ടിപ്പം.

Cleaning and Stock preparation. Paper/board making process, Finishing and

Cleaning and Stock preparation. Paper/board making process, Finishing and is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Cleaning and Stock preparation. Paper/board making process, Finishing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, cleaning and stock preparation. paper/board making process, finishing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Cleaning and Stock preparation. Paper/board making process, Finishing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-പ്ലേറ്റ് ക്ലീനിംഗ് ആൻഡ് സ്റ്റോക്ക് പ്രിപ്പറേഷൻ. പേപ്പർ/ബോർഡ്
മേക്കിംഗ് പ്രോസസ്സ്, ഫിനിഷിംഗ് ആൻഡ് പ്ലേറ്റ് ഓപ്പറേഷൻ പ്ലേറ്റ് definition/meaning
പ്ലേറ്റ് ഓപ്പറേഷൻ. പ്ലേറ്റ് ഓപ്പറേഷൻ പ്ലേറ്റ്, steps, application പ്ലേറ്റ് പ്ലേറ്റ്
പ്ലേറ്റ്. Technical terms English-പ്ലേറ്റ് പ്ലേറ്റ് പ്ലേറ്റ്.

3 Understanding Converting, Paper recycling

3 Understanding Converting, Paper recycling is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Converting, Paper recycling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding converting, paper recycling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Converting, Paper recycling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding Converting, Paper recycling
പേപ്പർ റീസൈക്കിൾ ചെയ്യുന്നതിന്റെ definition/meaning എന്താണ്. പേപ്പർ റീസൈക്കിൾ ചെയ്യുന്നതിന്റെ steps, application എന്താണ്. Technical terms English-ൽ എഴുതുക.
പേപ്പർ റീസൈക്കിൾ ചെയ്യുന്നതിന്റെ.

Paper and Board- Properties.

Paper and Board- Properties. is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Paper and Board- Properties. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, paper and board- properties. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Paper and Board- Properties. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പേപ്പർ Paper and Board- Properties. പേപ്പർ/ബോർഡ് definition/meaning പേപ്പർ/ബോർഡ്. പേപ്പർ/ബോർഡ് steps, application പേപ്പർ/ബോർഡ്. Technical terms English-പേപ്പർ/ബോർഡ്.

Paper Sizes. 1

Paper Sizes. 1 is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Paper Sizes. 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, paper sizes. 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Paper Sizes. 1 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പേപ്പർ Paper Sizes. 1 പേപ്പർ സൈസ് നിർവ്വചനം/definition/meaning പേപ്പർ സൈസ് നിർവ്വചനം. പേപ്പർ സൈസ് നിർവ്വചനം, steps, application പേപ്പർ സൈസ് നിർവ്വചനം പേപ്പർ സൈസ് നിർവ്വചനം. Technical terms English-പേപ്പർ സൈസ് നിർവ്വചനം.

Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for

Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, paper classifications and their uses. 1 importance of bis & tappi standards for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-പേപ്പർ Paper classifications and their uses. 1
Importance of BIS & TAPPI standards for പേപ്പർ definition/meaning
പേപ്പർ. പേപ്പർ പേപ്പർ, steps, application പേപ്പർ പേപ്പർ
പേപ്പർ. Technical terms English-പേപ്പർ പേപ്പർ.

1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting- Calendaring, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry.

1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting- Calendaring, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendaring, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 paper & its relation to printing industry. paper and board - introduction, invention of papermaking, raw materials used for paper and board making. pulping - definition, classifications-mechanical, chemical, semi - chemical and chemi - mechanical pulping. pulp washing, bleaching, refining, cleaning and stock preparation. paper/board making- overview on - manual method and machines used - cylinder press and fourdrinier machine. manufacturing process - forming, draining, pressing and drying. finishing and converting-calendaring, coating, slitting, cutting, packaging and labeling. mill conditioning. paper recycling-introduction. steps - sorting, deinking, pulping and papermaking. paper and board properties: define - printability & runnability properties. optical properties- color, gloss, opacity, metamerism, brightness, whiteness. structural properties- density, basis weight and grammage, caliper and bulk, wire and felt sides, moisture content and relative humidity, dimensional stability, grain

direction, porosity, smoothness. mechanical properties- tear resistance, tensile strength, bursting strength, folding endurance, stiffness, abrasion resistance. paper and board- sizes, classifications and their uses. importance of bis & tappi standards for paper & its relation to printing industry. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendarling, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendarling, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper

and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. പാലപ്പൊലിപ്പം പാലപ്പൊലിപ്പം definition/meaning പാലപ്പൊലിപ്പം. പാലപ്പൊലിപ്പം പാലപ്പൊലിപ്പം, steps, application പാലപ്പൊലിപ്പം പാലപ്പൊലിപ്പം പാലപ്പൊലിപ്പം. Technical terms English-പാലപ്പൊലിപ്പം പാലപ്പൊലിപ്പം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

packaging.

	Paper and board- Introduction, invention and	
M2.01		2
Understanding	raw materials used.	
	Pulping-Pulp Washing, Bleaching, Refining,	
		3
Understanding		
M2.02	Cleaning and Stock preparation.	
	Paper/board making process, Finishing and	
M2.03		3
Understanding	Converting, Paper recycling	
M2.04	Paper and Board- Properties.	1
Understanding		
M2.05	Paper Sizes.	1
M2.06	Paper classifications and their uses.	1
	Importance of BIS & TAPPI standards for	
M2.07		1
	paper & its relation to printing industry.	
	Series Test I	1

Contents:

Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 3

Explain about various aspects of plastics used in printing and packaging.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain about various aspects of plastics used in printing and packaging.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

2 Understanding classification

2 Understanding classification is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding classification using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding classification should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding classification means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-2 Understanding classification ഫലപ്രസാദം
ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps,
application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English-ഫലപ്രസാദം
ഫലപ്രസാദം.

Plastics-Manufacturing process-Extrusion.

Plastics-Manufacturing process-Extrusion. is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Plastics-Manufacturing process-Extrusion. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, plastics-manufacturing process-extrusion. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Plastics-Manufacturing process-Extrusion. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്ലാസ്റ്റിക്സ്-നിർമ്മാണ പ്രക്രിയ-എക്സ്ട്രൂഷൻ.

എക്സ്ട്രൂഷൻ എന്നതിന്റെ definition/meaning എഴുതുക. എക്സ്ട്രൂഷൻ ചെയ്യുന്നതിന്റെ steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Plastics used for printing and packaging.

Plastics used for printing and packaging. is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Plastics used for printing and packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, plastics used for printing and packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Plastics used for printing and packaging. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്ലാസ്റ്റിക്സ് പ്രിന്റിംഗ് ആൻഡ് പാക്കേജിംഗ്. പ്ലാസ്റ്റിക്സ് ന്റെ definition/meaning പ്ലാസ്റ്റിക്സ് ന്റെ ഉത്പാദനം, steps, application പ്ലാസ്റ്റിക്സ് ന്റെ ഉപയോഗം. Technical terms English-ൽ പ്ലാസ്റ്റിക്സ് പ്ലാസ്റ്റിക്സ്.

Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing,

Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing, is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, surface treatment on plastics 2 understanding chemical etching, waxing, varnishing, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്ലാസ്റ്റിക് സർഫസ് പ്രതിരോധം 2 Understanding Chemical Etching, Waxing, Varnishing, പ്ലാസ്റ്റിക് പ്രതിരോധം definition/meaning പ്ലാസ്റ്റിക് പ്രതിരോധം. പ്ലാസ്റ്റിക് പ്രതിരോധം പ്ലാസ്റ്റിക്, steps, application പ്ലാസ്റ്റിക് പ്രതിരോധം പ്ലാസ്റ്റിക്. Technical terms English-പ്ലാസ്റ്റിക് പ്രതിരോധം പ്ലാസ്റ്റിക്.

3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates.

3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding laminating, foiling on substrates. plastics - introduction. classification-natural, synthetic, semi-synthetic, thermosetting, thermoplastic. plastics - manufacturing process-extrusion basics. plastics used for printing and packaging - ldpe, hdpe, pp, ps, bopp, pvc, pet, cellophane, plastic corrugated board. surface treatment on plastics - surface cleaning, corona treatment, flame treatment, plasma treatment. chemical etching, waxing, varnishing, laminating, foiling on substrates. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Explain about various aspects of plastics used in printing and packaging.

M3.01	Plastic- definition, introduction & classification	2
M3.02	Plastics-Manufacturing process-Extrusion.	2
M3.03	Plastics used for printing and packaging.	1
M3.04	Surface treatment on plastics	2
M3.05	Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates.	3

Contents:
Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Describe about various aspects of miscellaneous printing materials.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe about various aspects of miscellaneous printing materials.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Dampening Solution and Blanket.

Dampening Solution and Blanket. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Dampening Solution and Blanket. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, dampening solution and blanket. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Dampening Solution and Blanket. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Dampening Solution and Blanket.

日本語から英語へ訳す definition/meaning 日本語から英語へ訳す。 日本語から英語へ訳す, steps, application 日本語から英語へ訳す 日本語から英語へ訳す。 Technical terms English-日本語 日本語から英語へ訳す。

Leather, Al foil, Glass.

Leather, Al foil, Glass. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Leather, Al foil, Glass. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, leather, al foil, glass. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Leather, Al foil, Glass. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പർവ്വതം Leather, Al foil, Glass. പർവ്വതം പർവ്വതം പർവ്വതം definition/meaning പർവ്വതം പർവ്വതം പർവ്വതം. പർവ്വതം പർവ്വതം പർവ്വതം, steps, application പർവ്വതം പർവ്വതം പർവ്വതം. Technical terms English-പർവ്വതം പർവ്വതം പർവ്വതം.

Adhesive.

Adhesive. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Adhesive. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, adhesive. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Adhesive. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പ്രതിരോധനം Adhesive. പ്രതിരോധനം എന്നതിന്റെ definition/meaning എന്താണ്. പ്രതിരോധനം എങ്ങനെ പ്രയോജനപ്പെടുന്നു, steps, application എങ്ങനെ പ്രയോജനപ്പെടുന്നു. Technical terms English-ൽ എങ്ങനെ പ്രയോജനപ്പെടുന്നു.

Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials &

Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials & is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials & using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, metals used in printing and packaging. 1 understanding photographic and light sensitive materials & should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials & means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇരുമ്പ് Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials & ഇരുമ്പ് നിർമ്മാണം definition/meaning ഇരുമ്പ് നിർമ്മാണം. ഇരുമ്പ് നിർമ്മാണം steps, application ഇരുമ്പ് നിർമ്മാണം ഇരുമ്പ്. Technical terms English-ഇരുമ്പ് നിർമ്മാണം.

2 Understanding Image carriers.

2 Understanding Image carriers. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Image carriers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding image carriers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Image carriers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-2 Understanding Image carriers.
 definition/meaning . steps, application . Technical terms English-
 .

Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das,Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1. <https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3. <https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5. <https://en.wikipedia.org/wiki/Plastic> 6. [https://en.wikipedia.org/wiki/Substrate_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7. <https://www.printweek.in/features/dampening-solutions> 8. <http://printwiki.org/Blanket>

Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das,Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook

of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1.
<https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3.
<https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5.
<https://en.wikipedia.org/wiki/Plastic> 6.
[https://en.wikipedia.org/wiki/Substrate_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7.
<https://www.printweek.in/features/dampening-solutions> 8.
<http://printwiki.org/Blanket> is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1.
<https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3.
<https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5.
<https://en.wikipedia.org/wiki/Plastic> 6.
[https://en.wikipedia.org/wiki/Substrate_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7.
<https://www.printweek.in/features/dampening-solutions> 8.
<http://printwiki.org/Blanket> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, plate treating solutions & cleaning solutions. 1 understanding dampening solution- purpose of use, ingredients and their properties. blanket - function and types. adhesive: types, properties and uses. leather - purpose of use, types. aluminum foil - properties and application. glass- introduction, types, properties and application. metals - types, characteristics and use. photographic and light sensitive materials - types, features and use. image carriers: cross - sectional view and imaging features of - lithographic plate, gravure cylinder, flexo plate, screen printing stencil, and digital imaging. plate treating solutions. cleaning solutions. text / reference: t/r book title/author t1 tulika das, chemistry in printing, 2nd edition, barnana prakashani, 2011. r.h. leach and r.j. pierce ed, the printing ink manual, 5th edition, kluer r1 academic publisher. bishwanath chakravarty, a hand book for printing and packaging, r2 galgotia publications pvt. ltd, 1997. gatf staff solving offset ink problems gatf, 1998 n.r.elred & t. scarlet, what the printer should know about ink, gatf, r3 1995 n.r.elred & t. scarlet, what the printer should know about paper, gatf, r4 1995 r5 n.r.elred & t. scarlet, chemistry for the graphic arts, gatf, 1992. r6 p. kipphan, handbook of print media, springer, 2002. online resources: sl.no website link 1. <https://en.wikipedia.org/wiki/ink> 2. <http://printwiki.org/ink> 3. <https://en.wikipedia.org/wiki/paper> 4. <http://printwiki.org/pulping> 5. <https://en.wikipedia.org/wiki/plastic> 6. [https://en.wikipedia.org/wiki/substrate_\(printing\)](https://en.wikipedia.org/wiki/substrate_(printing)) 7. <https://www.printweek.in/features/dampening-solutions> 8. <http://printwiki.org/blanket> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1. https://en.wikipedia.org/wiki/Ink 2. http://printwiki.org/Ink 3. https://en.wikipedia.org/wiki/Paper 4. http://printwiki.org/Pulping 5. https://en.wikipedia.org/wiki/Plastic 6. https://en.wikipedia.org/wiki/Substrate_(printing) 7. https://www.printweek.in/features/dampening-solutions 8. http://printwiki.org/Blanket means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1. <https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3. <https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5. <https://en.wikipedia.org/wiki/Plastic> 6. [https://en.wikipedia.org/wiki/Substrate_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7. <https://www.printweek.in/features/dampening-solutions> 8. <http://printwiki.org/Blanket> □□□□□□□□□□□□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□□□, steps, application □□□□□□ □□□□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Describe about various aspects of miscellaneous printing materials.

M4.01 Dampening Solution and Blanket. 2

Understanding

M4.02 Leather, Al foil, Glass. 2

Understanding

M4.03 Adhesive. 1

Understanding

M4.04 Metals used in printing and packaging. 1

Understanding

Photographic and light sensitive materials &
M4.05 2

Understanding

Image carriers.

M4.06 Plate treating solutions & Cleaning solutions. 1

Understanding

Series Test II 1

Contents:

Dampening solution- purpose of use, ingredients and their properties. Blanket - function

and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum

foil - properties and application. Glass- introduction, types, properties and application.

Metals - types, characteristics and use. Photographic and light sensitive materials - types,

features and use. Image Carriers: Cross - sectional view and imaging features of

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Ink-introduction and components. (3 marks)

Answer: Begin with the standard definition or identity of *Ink-introduction and components*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Ink properties. (3 marks)

Answer: Begin with the standard definition or identity of *Ink properties*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Ink manufacturing. (3 marks)

Answer: Begin with the standard definition or identity of *Ink manufacturing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Types of printing inks 3 **Understanding Ink-** Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink,

Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and. (3 marks)

Answer: Begin with the standard definition or identity of *Types of printing inks* 3 *Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: തിരിച്ചറയ്ക്കുന്നതിനായി നിർവ്വചനം, തത്വം/ഘട്ടം, പ്രയോഗം തുടങ്ങിയവ ഉൾപ്പെടെയുള്ളവ.

Module 2

Define or explain 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: തിരിച്ചറയ്ക്കുന്നതിനായി നിർവ്വചനം, തത്വം/ഘട്ടം, പ്രയോഗം തുടങ്ങിയവ ഉൾപ്പെടെയുള്ളവ.

□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Cleaning and Stock preparation. Paper/board making process, Finishing and. (3 marks)

Answer: Begin with the standard definition or identity of *Cleaning and Stock preparation. Paper/board making process, Finishing and.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain 3 Understanding Converting, Paper recycling. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Converting, Paper recycling.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Paper and Board- Properties.. (3 marks)

Answer: Begin with the standard definition or identity of *Paper and Board- Properties..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□.

Module 3

Define or explain 2 Understanding classification. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding classification*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Plastics-Manufacturing process-Extrusion.. (3 marks)

Answer: Begin with the standard definition or identity of *Plastics-Manufacturing process-Extrusion..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Plastics used for printing and packaging.. (3 marks)

Answer: Begin with the standard definition or identity of *Plastics used for printing and packaging..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing,. (3 marks)

Answer: Begin with the standard definition or identity of *Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing,.* State the governing principle,

list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Dampening Solution and Blanket.. (3 marks)

Answer: Begin with the standard definition or identity of *Dampening Solution and Blanket..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Leather, Al foil, Glass.. (3 marks)

Answer: Begin with the standard definition or identity of *Leather, Al foil, Glass..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Adhesive.. (3 marks)

Answer: Begin with the standard definition or identity of *Adhesive..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials &. (3 marks)

Answer: Begin with the standard definition or identity of *Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials &.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Ink-introduction and components**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding classification.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Dampening Solution and Blanket..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://en.wikipedia.org/wiki/Ink> <http://printwiki.org/Ink>
<https://en.wikipedia.org/wiki/Paper> <http://printwiki.org/Pulping>
<https://en.wikipedia.org/wiki/Plastic>
- [https://en.wikipedia.org/wiki/Substrate_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing))
- <https://www.printweek.in/features/dampening-solutions> <http://printwiki.org/Blanket>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3102. Verify institutional updates against the current official curriculum.